

❖ **WARNING:**
This preparation contains Sodium metabisulfite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulfites.

❖ **PREGNANCY AND LACTATION:**
Pregnancy
Bisoprolol should be given only if the potential benefit justifies the potential risk to the fetus Bisoprolol has pharmacological effects that may cause harmful effects on pregnancy and/or the fetus/newborn. In general, β-adrenoceptor blockers reduce placental perfusion, which has been associated with growth retardation, intrauterine death, abortion or early labor. Adverse effects (e.g. hypoglycemia and bradycardia) may occur in the fetus and newborn infant. If treatment with β-adrenoceptor blockers is necessary, β₁-selective adrenoceptor blockers are preferable. Bisoprolol should not be used during pregnancy unless clearly necessary. If treatment with Bisoprolol is considered necessary, the uteroplacental blood flow and the fetal growth should be monitored. In case of harmful effects on pregnancy or the fetus alternative treatment should be considered. The newborn infant must be closely monitored. Symptoms of hypoglycemia and bradycardia are generally to be expected within the first 3 days.

Lactation
It is not known whether this drug is excreted in human milk. Therefore, breastfeeding is not recommended during administration of Bisoprolol.

❖ **SIDE EFFECT:**
Cardiac disorders: AV-conduction disturbances, worsening of pre-existing heart failure, bradycardia.
Vascular disorders: Feeling of coldness or numbness in the extremities, hypotension, orthostatic hypotension.
Metabolism and nutrition disorders: Increased triglycerides.
Psychiatric disorders: Sleep disorders, depression, nightmares, hallucinations.
Nervous system disorders: Dizziness*, headache*, syncope
Eye disorders: Reduced tear flow (to be considered if the patient uses lenses), conjunctivitis.
Ear and labyrinth disorders: Hearing disorders.
Respiratory disorders: Bronchospasm in patients with bronchial asthma or a history of obstructive airways disease, allergic rhinitis.
Gastrointestinal disorders: Gastrointestinal complaints such as nausea, vomiting, diarrhea, constipation.
Hepatobiliary disorders: Increased liver enzymes (ALAT, ASAT), hepatitis.
Skin and subcutaneous tissue disorders: Hypersensitivity reactions (itching, flush, rash), beta-blockers may provoke or worsen psoriasis or induce psoriasis-like rash, alopecia.
Musculoskeletal and connective tissue disorders: Muscular weakness and cramps.
Reproductive system disorders: Potency disorders (reduced libido).
General disorders: Fatigue*, asthenia.
*These symptoms especially occur at the beginning of the therapy. They are generally mild and often disappear within 1-2 weeks.

❖ **CONTRAINDICATION:**
Bisoprolol must not be used in patients with:

- Acute heart failure or during episodes of heart failure decompensation requiring intravenous therapy with substances increasing the contractility of the heart
- Shock induced by disorders of cardiac function (cardiogenic shock)
- Severe disturbances of atrioventricular conduction (second or third degree AV block) without a pacemaker
- Sick sinus syndrome
- Sinoatrial block
- Slowed heart rate, causing symptoms (symptomatic bradycardia)
- decreased blood pressure, causing symptoms (symptomatic hypotension)
- severe bronchial asthma or severe chronic obstructive pulmonary disease
- severe forms of peripheral arterial occlusive disease or Raynaud syndrome
- untreated tumors of the adrenal gland (pheochromocytoma)
- metabolic acidosis
- hypersensitivity to Bisoprolol or to any of the excipients

❖ **DRUG INTERACTION:**
Combinations not recommended

- Calcium antagonists of the Verapamil type and to a lesser extent of the Diltiazem type: Negative influence on contractility and atrio-ventricular conduction. Intravenous administration of Verapamil in patients on beta-blocker treatment may lead to profound hypotension and atrioventricular block.
- Centrally acting antihypertensive drugs such as Clonidine and others (e.g. Methyldopa, Moxonidine, Rilmenidine): Concomitant use of centrally acting antihypertensive drugs may further decrease the central sympathetic tonus (reduction of heart rate and cardiac output, vasodilation). Abrupt withdrawal, particularly if prior to beta-blocker discontinuation, may increase risk of “rebound hypertension”.

Combinations to be used with caution

- Calcium antagonists of the dihydropyridine type such as Nifedipine: Concomitant use may increase the risk of hypotension, and an increase in the risk of a further deterioration of the ventricular pump function in patients with heart failure cannot be excluded.
- Class-I antiarrhythmic drugs (e.g. Disopyramide, Quinidine): Effect on atrio-ventricular conduction time may be potentiated and negative inotropic effect increased
- Class-III antiarrhythmic drugs (e.g. Amiodarone): Effect on atrio-ventricular conduction time may be potentiated.
- Topical beta-blockers (e.g. eye drops for glaucoma treatment) may add to the systemic effects of Bisoprolol.
- Parasympathomimetic drugs: Concomitant use may increase atrio-ventricular conduction time and the risk of bradycardia.
- Insulin and oral antidiabetic drugs: Intensification of blood sugar lowering effect. Blockade of beta-adrenoreceptors may mask symptoms of hypoglycemia.
- Anesthetic agents: Attenuation of the reflex tachycardia and increase of the risk of hypotension
- Digitalis glycosides: Reduction of heart rate, increase of atrio-ventricular conduction time.
- Non-steroidal anti-inflammatory drugs (NSAIDs): NSAIDs may reduce the hypotensive effect of Bisoprolol.
- Beta-sympathomimetic agents (e.g. Isoprenaline, Dobutamine): Combination with Bisoprolol may reduce the effect of both agents.
- Sympathomimetics that activate both beta- and alpha-adrenoceptors (e.g. noradrenaline, Adrenaline): Combination with Bisoprolol may unmask the alpha-adrenoceptor-mediated vasoconstrictor effects of these agents leading to blood pressure increase and exacerbated intermittent claudication. Such interactions are considered to be more likely with nonselective beta-blockers. Higher doses of adrenaline may be necessary for treatment of allergic reactions.
- Concomitant use with antihypertensive agents as well as with other drugs with blood pressure lowering potential (e.g. tricyclic antidepressants, barbiturates, phenothiazines) may increase the risk of hypotension.
- Moxisylyte: Possibly causes severe postural hypotension.

Combinations to be considered

- Mefloquine: Increased risk of bradycardia
- Monoamine oxidase inhibitors (except MAO-β inhibitors): Enhanced hypotensive effect of the beta-blockers but also risk for hypertensive crisis.

❖ **OVERDOSE AND TREATMENT:**
Symptoms
The most frequent signs of Bisoprolol overdose include slow heart rate (bradycardia), marked drop in blood pressure, acute heart failure, hypoglycemia and bronchospasm.

Treatment
In general, if overdose occurs, Bisoprolol treatment is stopped and supportive and symptomatic treatment is provided.

❖ **STORAGE:**
Store at temperature of not more than 30°C.

❖ **DOSAGE FORM AND PACKAGING AVAILABLE:**
2.5 mg Tablet, Blister 10x10's
5 mg Tablet, Blister 10x10's

❖ **DATE OF REVISION:**
April 24, 2024

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomchojorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

BACK

16.5 cm.

29.5 cm.

Product	Code No.	Dimension	Packaging Type	Thickness
BISLOC 2.5-5	IBMY 0042	W 29.5 x L 16.5 cm.	Wood Free Paper (กระดาษไม้นึก)	60 g (0.08 mm.)
PM Specification				
Designed by: PLC: (Date)		Checked by: IRA: (Date)		Approved by: QCC-PM: (Date) MD (Only Pattern and Coding) (Date) CUSTOMER/SALES DEPT: (Date)